

2.2.7 DICOM

HP-Z0000YPR-101005-02-012

DICOM allows you to specify the parameters for the DICOM communication interfaces for your system and for the remote systems.

General remarks

In this section, you define the Application Entity Titles for the local system and general time-out settings for all DICOM processes.

Step 1: General - Local AETs

In this section, you define the Application Entity Title for the local system. Since 3 different Windows NT servers (processes) are running, 3 AEs are also available by default.

NOTE

The default AET name is the computer name. Changing the computer name could also change the AETs. A pop-up window lets you select whether or not the AET should also be changed.

Definition of the Application Entity Title

According to Part 8 of the DICOM Standard, the AET identifies a UNIQUE DICOM Service or application on a specific system in the NETWORK. The name consists of characters, excluding the control characters LF, FF, CR, and ESC. The AET is limited to a length of 16 bytes.

It is permitted to use the same AET for different DICOM services, e.g., Storage, Basic Print, Modality Worklist, etc. However, you cannot use the same AET twice for the same DICOM service.

Some configuration tools in Siemens products allow the same AET to be entered twice for the same service. In this case, the Merge tool kit will always select the destination of the first AET.

All Siemens products that support both DICOM and the PACSnet protocol use the PLA prefix (first 10 characters of the PACSnet Logical Address) and a 4-character suffix, e.g., 049SA1DC01DRSP.

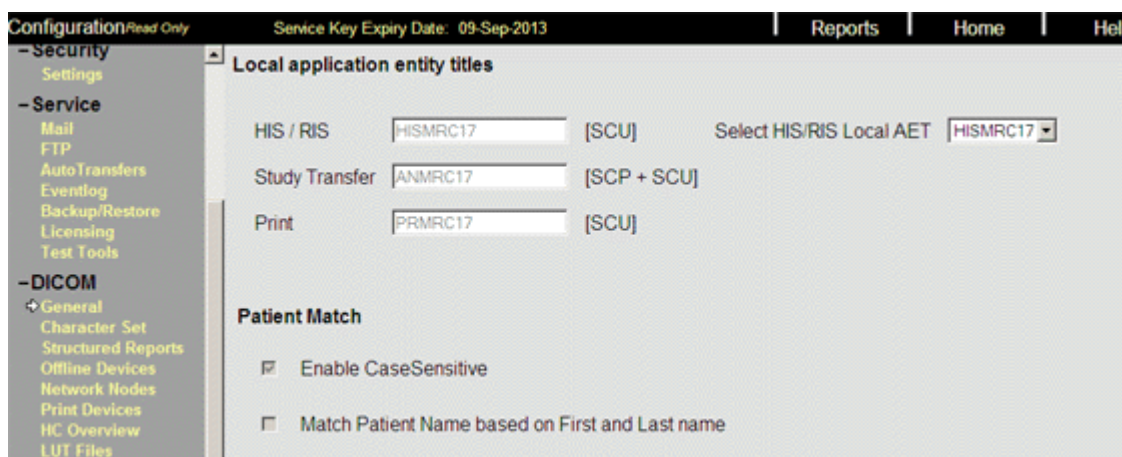


Fig. 32: Configuration - DICOM - General

- **HIS/RIS Server**

This server performs the DICOM Modality Worklist Services. Usually, this AET needs to be known at the RIS server as well as for authentication purposes. The default AET is the HOST NAME (upper case). Please be aware that changing the local host name may also change the AET.

- **Study Transfer Server**

This server performs the Storage and Query/Retrieve DICOM Services. Usually, this AET needs to be known at the remote nodes for authentication purposes. The default AET is the HOST NAME (upper case). Please be aware that changing the local host name may also change the AET.

- **Print server**

This server performs the Basic Print Service. Usually, this AET needs to be known at the camera as well as for authentication purposes. The default AET is the HOST NAME (upper case). Please be aware that changing the local host name may also change the AET.

NOTE

The port number of the Study Transfer Provider (SCP) is 104. The port numbers of the HIS/RIS and PRINT SCUs are not relevant because they change with each association and are not required for configuration.

Step 2: General DICOM Settings

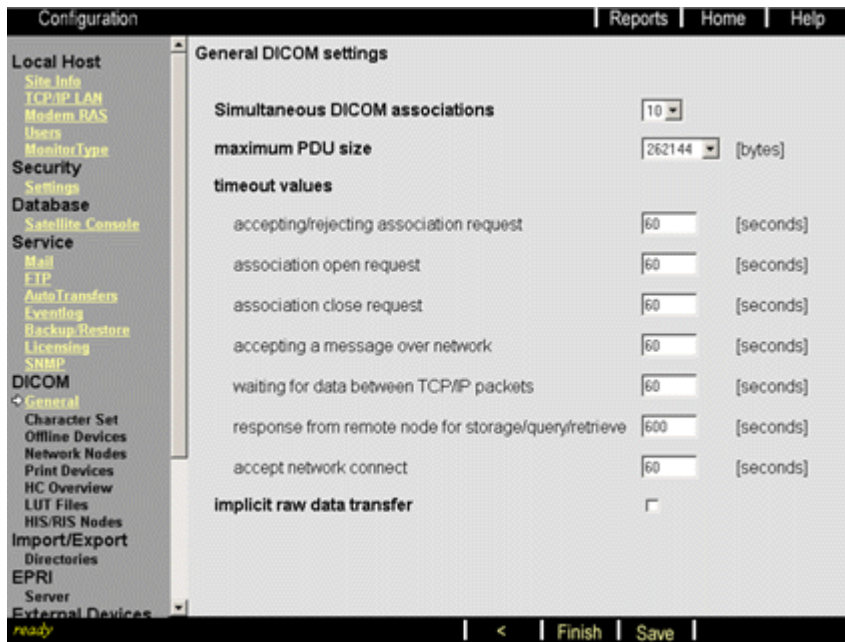


Fig. 33: General DICOM settings menu

- **Simultaneous DICOM Associations**

Set the number of DICOM associations to be handled simultaneously. Reduce the number if there are performance problems.

- **Maximum PDU Size**

The Data Unit Packet size defines how many bytes of data are collected at the interface between TCP/IP and DICOM. The higher the value of the PDU size, the better the performance. During the association negotiation at the beginning of each DICOM session, the maximum PDU size is negotiated between the two involved systems - the lower one is used.

Timeout Values

- **Accepting/Rejecting Association Request**

The ARTIM_TIMEOUT is set in the mergecom.pro file and is the time out to wait for an association request or to wait for a peer to shut down an association.

- **Association Open Request**

The ASSOC_REPLY_TIMEOUT is set in the mergecom.pro file and is the time out to wait for a reply to an association request.

- **Association Close Request**

The RELEASE_TIMEOUT is set in the mergecom.pro file and is the time out to wait for a reply to an association release.

- **Accepting a Message over Network**

The WRITE_TIMEOUT is set in the mergecom.pro file and is the time out to wait for a network write to be accepted.

- **Waiting for Data between TCP/IP Packets**

This INACTIVITY_TIMEOUT is active during the message transfer (e.g. image) and needs to be increased when performance problems in the network occur (switches, routers).

- **Response from Remote Node for Storages' Query/Retrieve**

This is the time out between an C_MOVE request (Search - Import) and the import (retrieve) of the selected images.

- **Accept Network Connect**

This CONNECT_TIMEOUT is set in the mergecom.pro file and is the time out to wait for a network connect to be accepted.

- **Implicit Raw Data Transfer**

Allows general transfer of raw data to a remote DICOM destination. In addition, it is necessary to select 'raw data' in the customer user interface when transferring images. The raw data is sent in private image header attributes, resulting in a very large image header.

Character set

In this menu, you define the character set which is used when a new image or object is created on the local system. As soon as images or objects are imported (Storage, Query/Retrieve, Worklist) to the system, the character set of the imported objects is used.



Fig. 34: Character Set menu

DICOM Default Character Sets

Depending on the country and the institutions environment, it offers a selection of different standard character sets:

- **Default Repertoire, ISO_IR6:** The most popular character set is the 7-bit ASCII Code. The registry entry of ASCII in ISO 2375 is ISO_IR 6, which is used as the defined term in the DICOM Standard for the 'Specific Character Set' attribute 0008,0005 (although DICOM defines just a subset of ISO_IR 6).
- **Latin Alphabet No.1, ISO_IR100:** The 8-bit character sets are defined in ISO 8859, which is a series of more than 10 standardized multilingual graphic character sets like:
 - Latin1 (West European) **ISO_IR 100**
 - Latin2 (East European) **ISO_IR 101**
 - Latin3 (South European) **ISO_IR 109**
 - Latin4 (North European) **ISO_IR 110**
 - Cyrillic **ISO_IR 144**
 - Arabic **ISO_IR 127**
 - Greek **ISO_IR 126**
 - Hebrew **ISO_IR 138**
 - Latin5 (Turkish) **ISO_IR 148**
- **Japanese, Katakana, ISO_IR13:** The Japanese character set description is composed of two character sets, ISO_IR 14 (Japanese version for Roman Characters) and ISO_IR 13 (Katakana character set). More detailed information can be found in the DICOM Standard, Section 5, Appendix H, and Section 3 C.12, General Modules.
- **Unicode in UTF-8, ISO_IR192**
Unicode is the common stand and supports all other DICOM supported character sets, except GB18030. The option character set extension is not possible.
- **Chinese, GB18030**
Used for chinese language. This character set is even larger and includes the Unicode character set. The option character set extension is not possible.

Allow character set extension

DICOM allows you to use more than one character set in an object. If you want to enter Swedish characters and Russian characters, for example, you need to switch from one code to the other in one string by adding a certain Escape Sequence. What is referred to as this 'code extension' technique is defined in ISO 2022. As a result, the defined term for the 'Specific Character Set' attribute 0008,0005 receives the prefix ISO 2022 to show the application that code extension is being used (e.g., ISO 2022 IR 100).

With this prefix, the application is prepared for decoding the Escape Sequences. Otherwise, it can fail when an Escape Sequence appears.

Structured Reports

A configuration interface is provided to offer the possibility to assign a certain viewing application to a certain Structured Report type of e.g. another vendor. The Report type is specified by the Template ID and Mapping Resource as part of the DICOM header of an image, whereas the assigned Siemens viewer for that system is already predefined.

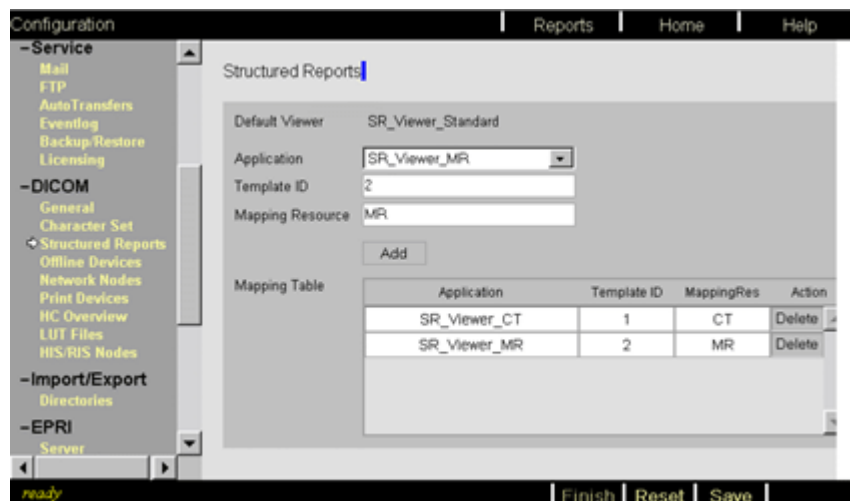


Fig. 35: Configuration Structure of Report View

Default Viewer

The default viewer is pre-configured and can not be changed.

Application

It provides a list of installed and known viewing applications on the system to select the proper viewer for a Structured Report type.

Template ID

DICOM header information to specify the Structured Report type together with the **Mapping Resource**.

Mapping Table

It shows the list of configured combinations of viewing application and concerned Template ID / Mapping Resource.

A new combination can be inserted into the table by selecting the viewing application out of the **Application** select box and entering the concerning **Template ID** and **Mapping Resource**.

The insert into the table is executed with the **Add** button and is permanently stored with **Save**.

A complete table row (stands for a viewing application and report type combination) can be deleted by the **Delete** button behind each row.

Offline devices

In this section, the settings for the DICOM offline devices (Media Storage) on the local host are defined.

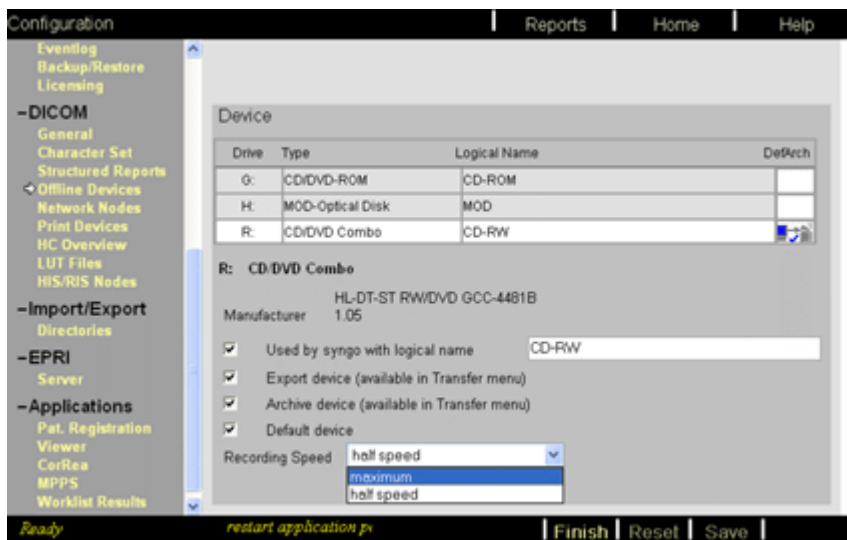


Fig. 36: DICOM - Offline Devices menu

The available Media Storage devices of the system are automatically detected by syngo and shown in the device table with the assigned drive letter.

The particular device is selected by mouse click.

Device Type



Fig. 37: Offline Devices Type

Due to the limitation of the operating system (Removable Media Abstraction Layer) the system is only able to detect devices of type CD/DVD Combo. Therefore those devices are recognized and no combo box are offered for changing the device type. Others have to be allocated to the proper device type.

- **CD/DVD Combo**

Select for combined CD and DVD Burner.

- **CD-R**

Select for CD Burner only.

- **CD/DVD-ROM**

Select for CD and DVD Reader only, without writing capability.

- **MOD-Optical Disk**

Select if the device is a MOD (Only supported by syngo MR upgrade systems from NUMARIS 3 / 3.5).

NOTE

Please be aware that syngo might not distinguish between CD-ROM and CD-RW.

Insert a CD to verify if you configure the correct drive letter for the corresponding drive!

Special syngo device settings

- **Used by syngo with logical name**

Enables the device to be used in syngo. If enabled an **unique** logical name has to be given (free text).

- **Export Device**

Defines if the device shall be used as export device within syngo.

NOTE

It's limited to the device types CD-R and CD/DVD Combo only, which can be selected for export.

- **Archive device**

This check box is available depending from the system type (modality). It defines if the device shall be used as archive device.

NOTE

Only one CD-R or CD/DVD Combo drive can be defined as archive device.

- **Default device**

Defines if the device shall be used as default device. Only one device per system can be defined. The defined device is marked as default within the device table by an icon in the last column (DefArch).

It's not applicable for CD-R drives.

Recording Speed

The recording speed of the selected device can be configured for half or maximum speed.

See following table for recommended logical names:

Drive	Type	Logical Name
R:	CD	CD-ROM or DVD-ROM
S:	CD-R or CD/DVD Combo	CD-RW or DVD-RW

Network nodes

Definition of DICOM nodes for Storage (send and receive) and Query/Retrieve. Please note that DICOM printers and the Radiology Information System (RIS) are not configured here.

The Image transfer or Query/Retrieve targets entered can be selected in the “Transfer - Send to...” menu of the viewer or patient browser.

The Query/Retrieve targets entered may be selected in the viewer or patient browser menus “Patient - Search”.

NOTE

DICOM nodes that send only to the local system need to be configured as well, since their IP addresses will be checked. One problem is that these nodes are also visible in the viewer or patient browsers “Transfer - Send to...” menu as a send destination. To prevent this, enter just the TCP/IP settings (Step 1) and skip the DICOM settings (Step 2).

Required Data

It is the responsibility of the Project Manager to obtain the following data from the customer.

- Host name and TCP/IP address of the remote host.
- Application Entity Title (AET) and port number of the corresponding DICOM processes on the remote host
- Transfer syntax (JPEG, Explicit Big Endian, ...)
- Which DICOM service and in which direction (SCU, SCP)
- Usually the remote systems need to be configured as well. It is recommended that the service engineer from the other vendor be on-site at the same time.

Step 1: TCP/IP Settings for DICOM Network Nodes

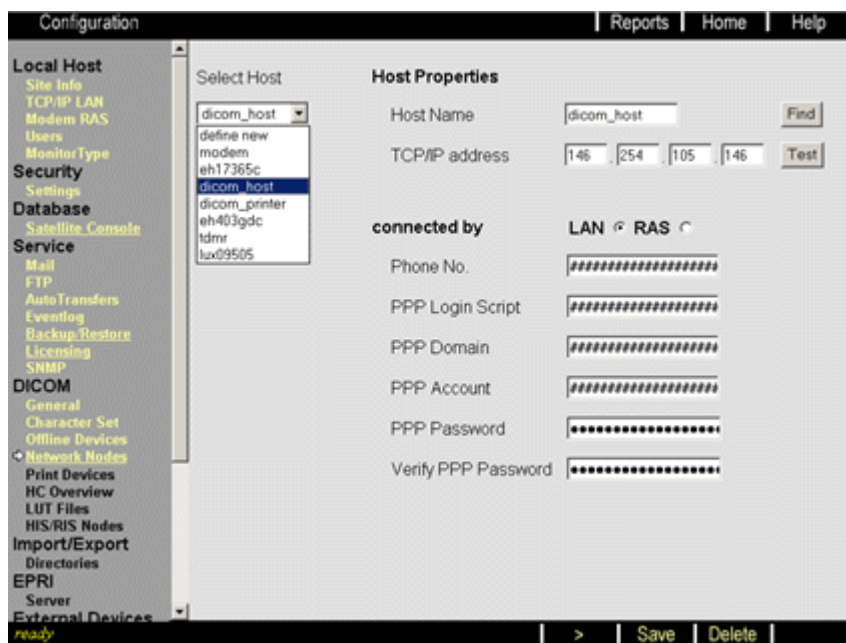


Fig. 38: DICOM Network Nodes menu - step 1

Select Host

Click on the “Select Host” drop-down menu to see whether the DICOM node is already listed.

If the DICOM node is not listed, you need to enter the host name and the IP address.

Host Properties

- Host Name**

Enter the host name of the DICOM node.

NOTE**Restrictions for the Host Name:**

The host name may contain up to 15 characters. The characters “_” and “-” should be avoided, since they could cause problems (even though it is possible to enter them). The computer name may not consist of just numeric characters or start with a numeric character.

- Find**

Select the **Find** button to check whether the host name is known at the WINS or Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.

NOTE

Find works for RAS only if the RAS connection has been established previously.

- **TCP/IP Address**

If no DNS is used, you need to manually enter the IP address into these boxes.

RAS requires two IP addresses - one for the local system and the second for the remote system. For RAS, you need to enter the second IP address.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address.

A TCP/IP ping may fail for the following reasons:

- The remote host is turned off, doesn't answer on ping or is not in the same network (or sub net), or the gateway is not configured in the routing table.
- Any networking device (router, switch, bridge,...) denies access to the target network.

Connected by LAN / RAS

If the DICOM node is connected via Ethernet (including a router), you need to select the **LAN** button.

NOTE

(RAS) Modems are not supported by syngo MR systems.

Step 2: DICOM Settings for DICOM Nodes

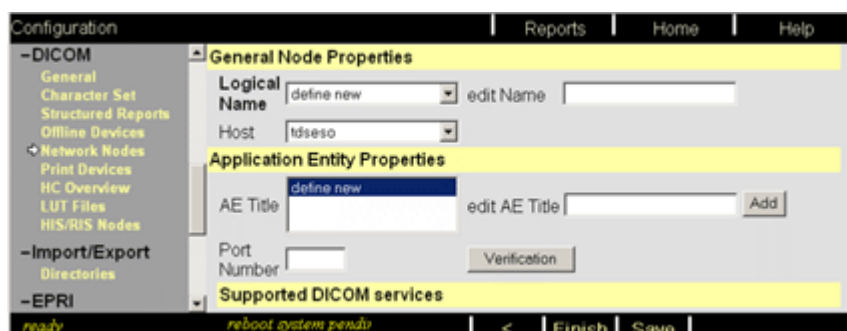


Fig. 39: DICOM Network Nodes Menu - step 2

General Node Properties

- **Logical Name**

Select the logical name for the DICOM device or define a new one in the 'Edit Name' box. The logical name is displayed in the customer user interface at "Transfer - Send to..." for Storage and at "Patient - Search" for Query/Retrieve.

- **Edit Name**

Create or update a logical name for the DICOM device.

- **Host**

Select the host name (computer name) of the DICOM device.

Application Entity Properties

- **AE Title**

Select the Application Entity Title for the DICOM device or define a new one in the “Edit AE Title” box. The AETs already configured are shown in this pull-down menu.

NOTE

Restrictions for the AET:

- **Maximum 16 characters are allowed**
 - **BLANKS and " (double quotes) are not allowed**
 - **You may use alphabetic and numeric characters as well as “ _ “ and “-“.**
 - **It is recommended to write AETs in upper case.**
-

- **Edit AE Title**

Enter the Application Entity Title of the DICOM device.

- **Port Number**

Enter the Application Entity Title and port number of the DICOM device and click on the Add button. The port number is the TCP/IP port where the DICOM process (defined by the AET) at the DICOM device is listening.

- **Add / Del**

When selecting a previously configured AET, the Add button will change to the Del button to delete this AET.

- **Verification**

DICOM verification is a DICOM service for test purposes. What is called a C-ECHO command is transferred to the selected DICOM service, which is defined by the IP address, the port number, and the AET.

A C-ECHO may fail for the following reasons:

- TCP/IP address, AET, or port number is not configured correctly.
- The DICOM process at the remote host is not running.
- Certain products check the IP address or AET of the sending system. If the AET or IP address of the local system is not entered correctly there, DICOM verification will fail.
- The remote host does not support DICOM Verification as a Service Class Provider (see DICOM Conformance Statement).

Supported DICOM Services

Select which DICOM services are to be performed by the particular system.

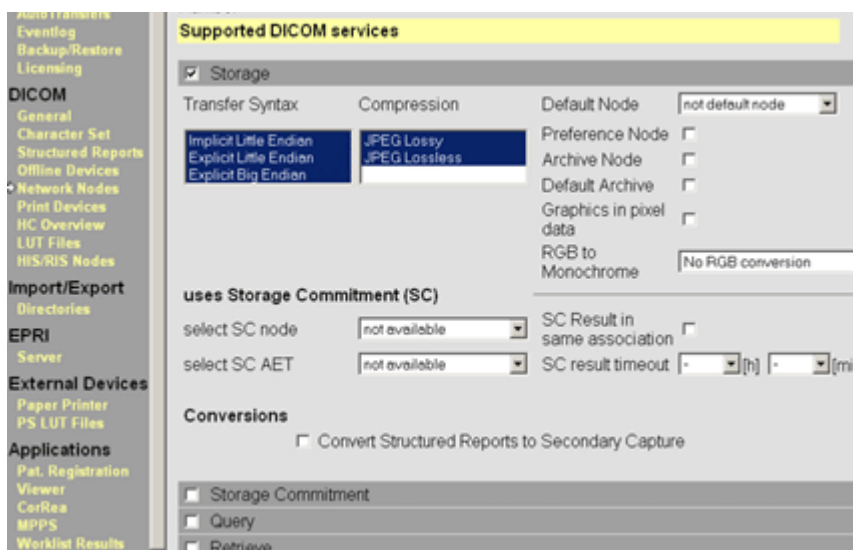


Fig. 40:

Storage

• Transfer Syntax / Compression

Select the transfer syntaxes proposed by the local system to the remote DICOM node during the association negotiation. When sending images to the DICOM node, this node selects one of the proposed transfer syntaxes as the result transfer syntax.

NOTICE

This setting has no effect when receiving images.

⇒ Syngo will accept all transfer syntaxes, regardless of what is selected in this menu.

NOTICE

How can you deselect both compression transfer syntaxes from the selection box?

⇒ Just click on one of them to remove it. The second one can be removed by also selecting the CTRL button.

- **Implicit Little Endian:** Implicit (without) value representation (string, integer, date, ...). The value representation for each attribute will not be sent in the image header and needs to be added at the destination using a data dictionary file.
Little Endian is a synonym for sending the least significant bit first (byte swap).
- **Explicit Little Endian:** Explicit (with) value representation (string, integer, date, ...). The value representation for each attribute is sent in the image header.
Little Endian is a synonym for sending the least significant bit first (byte swap).
- **Explicit Big Endian:** Explicit (with) value representation (string, integer, date, ...). The value representation for each attribute is sent in the image header.
Big Endian is a synonym for sending the most significant bit first (byte swap).

- **JPEG Lossy:** JPEG defines a set of compression algorithms. With Lossy JPEG, the transferred image will lose data. The quality factor can be selected in the customer user interface; this affects how much information will be lost. This transfer syntax should be selected only if required by the customer. It will usually be used for WAN (modem, ISDN) when the destination node does not need the full image quality.
- **JPEG Lossless:** This transfer syntax results in a much lower compression factor, but no data will be lost.
- **Default Node**
Define the order for display in the transfer menu by selecting the “first default” / “second default” or “no default” entry. Only the 1st and 2nd nodes will be displayed in the customer user interface under “Transfer - Send to ...”. For the other nodes, an additional pull-down menu needs to be opened in the customer user interface.
- **Preference Node**
One DICOM storage device may be selected as the preference node. This affects only the keyboard settings; when you click on the + button, the selected study is transferred to this preference node.
- **Archive Node**
If you check this box, the studies sent to the DICOM device will receive an “Archive flag” in the patient browser in the customer user interface.
- **Default Archive**
If you check this box, this DICOM device is set as the Archive device in the customer user interface.
- **Graphics in Pixel Data**
By default, syngo-based systems write the overlay data (annotations, distance, angle, ROI, ...) to the attribute 6000.3000 and also as private attributes (group 0029). The private attributes can be used only by other syngo-based systems. They make it possible to reanimate the overlays (change them on the destination).

Several applications from other vendors are not able to read the overlays in 6000.3000. In this case, you need to select this button. The overlay data is then shifted in the pixel data to bit 12. If there are images using more than 12 bits, this shift cannot be performed.
- **RGB to Monochrome**
It can be configured that an RGB color image will be converted and sent as a monochrome image.
 - **No RGB Conversion**
All images with photometric information (RGB) are sent without color conversion
 - **Convert RGB Color Images**
If certain Ultrasound specific DICOM attribute are recognized, **but no color data present**, the image is converted from 3x8 bit RGB into 8 bit gray scale value and certain attributes are changed before sending. In addition all private attributes are removed
 - **Convert Ultrasound Non-Color Data**
If certain Ultrasound specific DICOM attribute are recognized the image is converted from 3x8 bit RGB into 8 bit gray scale value and certain attributes are changed before sending. In addition all private attributes are removed

Conversions

The check box provides the ability to configure node-specific whether DICOM Structured Report or DICOM Secondary Capture is supported. In this way it's be possible to send DICOM Secondary Capture results and reports to nodes that don't support DICOM Structured Report. Per default convert to Secondary Capture is unchecked.

Uses Storage Commitment

Here you need to define the Storage Commitment provider to whom the commitment is to be sent after the image transfer (Storage). It is not necessarily the same device as the storage destination, e.g., you can send the images to a workstation which automatically forwards them to an archive. In this case, you may send the Storage Commitment request to the archive device and not to the workstation.

- **Select SC Node**

Select the host name of the Storage Commitment provider.

- **Select SC AET**

Select the Application Entity Title of the Storage Commitment provider.

- **SC Result in Same Association**

If this box is selected, the Storage Commitment association will stay open until the response from the provider is received or the "SC result time out" is exceeded.

- **SC Result Timeout**

The Time out for Storage Commitment response applies both to results in the same or in a new association.

Storage commitment

Select this box to configure the appropriate DICOM device as Storage Commitment Provider (and supports this DICOM Service). It is not necessarily the Storage Commitment provider used after the image transfer from the local device (see "Uses Storage Commitment").

Query

Selecting this box assumes that the AET shown is a Query provider. The logical name will be displayed in the customer user interface in "Patient - Search". The Query service results in a list of the selected patients / studies of the DICOM node just configured.

Retrieve

To be able to retrieve (get) the selected images from the DICOM node, it is also necessary to select the "Retrieve" box. Usually, Query and Retrieve need to be selected.

The retrieve command initiates a Storage service at the remote DICOM node to send the images to the local system.

Print Devices

This section defines the DICOM laser cameras connected to your system via DICOM Basic Print. A postscript paper printer is not a DICOM printer! This system does not support the SPCI/SPDI camera interface.

Required Data

It is the responsibility of the Project Manager to obtain the required data from the customer.

- Host name and TCP/IP address of the remote host.
- Application Entity Title (AET) and port number of the corresponding DICOM processes on the remote host
- Usually, it is necessary for the camera to be configured as well. It is recommended that the service engineer from the printer vendor be on-site at the same time.

NOTE

If you need to delete a print device, the correct procedure is to start with menu 3 to menu 1. Otherwise, problems may result with the filming tab card in the user interface, registry settings, etc.

Step 1: TCP/IP Settings for DICOM Camera

Fig. 41: DICOM Print Devices - step 1

Select Host

Click on the “Select Host” drop-down menu to see whether the DICOM camera is already listed. If so, and if the related data are correct (including RAS settings), you are done and may continue with the next menu.

If the DICOM camera is not listed, you need to enter the host name and the IP address.

Host Properties

• Host Name

Enter the host name of the DICOM camera.

NOTE**Restrictions for the Host Name:**

The host name may contain up to 15 characters. The characters “_” and “-” should be avoided, since they could cause problems (even though it is possible to enter them). The computer name may not consist of just numeric characters or start with a numeric character.

- **Find**

Select the **Find** button to check whether the host name is known at the WINS or Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.

Find works for RAS only if the RAS connection has been established previously.

- **TCP/IP Address**

If no DNS is used, you need to manually enter the IP address into these boxes.

RAS requires two IP addresses - one for the local system and the second for the remote system. For RAS, you need to enter the second IP address here.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address.

Test works for RAS only if the RAS connection has been established previously. The first ping may fail because of a time out, since it takes time to establish the line.

A TCP/IP ping may fail for the following reasons:

- The remote host is turned off or not in the same network (or sub net), or the gateway is not configured.
- The TCP/IP address is configured incorrectly.
- Any networking device (router, switch, bridge,...) denies access to the other network.

Connected by LAN / RAS

For performance reasons, a DICOM laser camera should not be connected via RAS.

Step 2: DICOM Settings for DICOM Camera

Fig. 42: DICOM Print Devices - step 2

General Node Properties

- **Logical Name**

Select the logical name for the DICOM camera or define a new one in the “edit Name” box. Unlike “network nodes” and “HIS/RIS nodes”, the logical name is not visible at the customer user interface. It is used only as a link to the next menu (step 3).

- **Edit Name**

Create or update a logical name for the DICOM camera.

- **Host**

Select the host name (computer name) of the DICOM camera.

Application Entity Properties

- **AE Title**

Select the Application Entity Title for the DICOM camera or define a new one in the field “edit AE Title”. The AETs already configured are shown in this pull-down menu.

NOTE**Restrictions for the AET:**

- Maximum 16 characters are allowed
- BLANKS and " (double quotes) are not allowed
- You may use alphabetic and numeric characters as well as “_” and “-”.
- It is recommended to write AETs in upper case.

- **Edit AE Title**

Enter the Application Entity Title of the DICOM camera.

- **Port Number**

Enter the port number of the DICOM camera and click on the Add button.

The port number is the TCP/IP port where the DICOM process (defined by the AET) at the DICOM camera is listening.

- **Add / Del**

If a previously configured AET is selected, the **Add** button changes to the **Del** button, allowing you to delete this AET.

- **Verification**

DICOM verification is a DICOM service for test purposes. What is called a C-ECHO command is transferred to the selected DICOM service, which is defined by the IP address, the port number, and the AET.

A C-ECHO may fail for the following reasons:

- TCP/IP address, AET, or port number is not configured correctly.
- The DICOM process at the remote host is not running.
- Certain products check the IP address or AET of the sending system. If the AET or IP address of the local system is not entered correctly there, DICOM verification will fail.
- The remote host does not support DICOM Verification as a Service Class Provider (see DICOM Conformance Statement).

Step 3: Camera Parameter Settings

The screenshot shows the 'Configuration' window for 'DICOM Print Devices' at step 3. The left sidebar contains a tree view with categories like Local Host, Site Info, TCP/IP LAN, Modem RAS, Users, MonitorType, Security, Settings, Database, Satellite Console, Service, Mail, ETP, AutoTransfers, Eventlog, Backup/Restore, Licensing, SNMP, DICOM, General, Character Set, Offline Devices, Network Nodes, Print Devices, HC Overview, LUT Files, HIS/RIS Nodes, Import/Export, Directories, EPRI, Server, and External Devices. The main content area is titled 'Configuration' and has tabs for 'Reports', 'Home', and 'Help'. It contains several sections: 'Select HC device' with a 'define new' button; 'General Settings' with fields for Type (DICOM printer), Class (egfedrystar2000), DICOM Node, Location, and Comment (No Comment); 'Filing Properties' with fields for Hold printed filmjobs (10), Pixel Size (85 x 85), Film Sheet Formats Portrait (define new), Film Sheet Formats Landscape (define new), Medium type (blue film), and Film destination (processor). A 'Finish' button is at the bottom right.

Fig. 43: DICOM Print Devices - step 3

- **Select HC Device**

Previously configured DICOM cameras may be selected in this pull-down menu.

- **HC Device**

Enter the name of the camera that is displayed in the customer user interface on the Print tab card.

- **Default Camera**

If this box is selected, the camera will be set as the default camera. The other cameras need to be selected in a pull-down menu on the Print tab card in the customer user interface.

General Settings

- **Type**

You may select from the following camera types:

- **DICOM printer** : Select this item for all DICOM cameras and for SPDI/SPCI cameras connected via a DICOM interface box.
- **SPCI printer** : This item was originally reserved for a DICOM interface box from Siemens to SPDI/SPCI. This box is not released; as a result, this menu item is obsolete.

- **Class**

A pull-down menu with the released DICOM cameras is provided. Select your camera. Call Service if your camera is not available in the list.

- **DICOM Node**

The logical names defined in Step 2 of the camera configuration are displayed in this pull-down menu. Select the appropriate camera.

- **Location**

Enter the location of the camera. This information is displayed only in this configuration menu.

- **Comment**

Enter comments such as the help desk phone number of the camera vendor. This information is displayed only in this configuration menu.

Filming Properties

Fig. 44: DICOM Print Devices - step 3

- **Hold printed Film Jobs**

Even if the film has been printed successfully, the most recent films remain in the print queue. The number of films in the queue may be defined here.

- **Min. Density**

Set the minimum density value for the camera. The maximum density value is modality-dependent and can be set in the “LUT files” menu.

- **Pixel Size [1/1000 mm]**

This value is entered automatically, depending on the selected camera (class). This value may be modified here.

- **Film Sheet Formats**

Portrait / Landscape The default film sheet formats are displayed, depending on the selected camera (class). They may be modified or additional formats may be defined.

- **Number of Pixels [rows, columns]**

The formats are automatically filled in, depending on the selected camera (class). In the boxes below, the number of pixels (row and column) may be modified depending on the medium sheet size.

- **Medium Type**

Depending on the film used, select between “blue film”, “clear film”, and “paper”.

- **Film Destination**

You may select between “magazine” and “processor”. If you select magazine, the exposed film sheet needs to be developed in a darkroom. A processor will automatically develop the exposed film sheet.

- **Color Appearance**

You can select between monochrome (black & white) or color. This setting is necessary if you want to print color images (Ultrasound) to a camera that supports color.

- **24 bit Color:**

In this case, the camera negotiates the “Basic Color Print Management” and transfers the pixel data in RGB photometric interpretation.

- **8 bit Gray Scale:**

In this case, the camera negotiates the “Basic Gray Scale Print Management” and transfers the pixel data in MONOCHROME2 photometric interpretation with 8-bit pixel depth.

- **12 bit Gray Scale:**

In this case, the camera negotiates the “Basic Gray Scale Print Management” and transfers the pixel data in MONOCHROME2 photometric interpretation with 12-bit pixel depth (16 bit allocated for each pixel).

- **Background**

The background of the film may be set to black or white.

- **Transformation**

This value is sent to the camera if pixel interpolation is necessary. Usually the entire film sheet is calculated at the local system and recalculation is therefore not necessary at the camera. As a result, this parameter has no effect on the camera.

But be aware that an incorrect setting may not be accepted by a camera.

Nevertheless there are more and more high-resolution cameras available, whereby syngo sends only e.g. the half pixel matrix (each second pixel value). In this case the camera has to interpolate.

The following interpolation algorithms are possible:

Replicate: The interpolated value is identical to the last pixel value

Bilinear: The interpolated value (linear) is between the last pixel value and the next pixel value.

Cubic: The interpolated value (curve) is in between the last pixel value and the next pixel value.

No Magnification: Leaves the image as is.

HC Overview

The “Hard Copy Overview” menu allows the customer to adjust and test certain camera settings. There is no need for the field service engineer to use this menu, since the settings and tests can also be performed in the “hardcopy device” menu.



Fig. 45: DICOM HC Overview menu

These menu items are already described in the “hardcopy devices” menu.

LUT Files

This menu makes it possible to import, remove, and select LUT files (Look Up Table) for DICOM Basic Print. Currently, no tool is available for modifying a lookup table (LUT editor).

Step 1: Import of additional LUT Files

NOTE

Normally it's not required to import additional LUT's, click > to select the syngo MR LUT for your console.

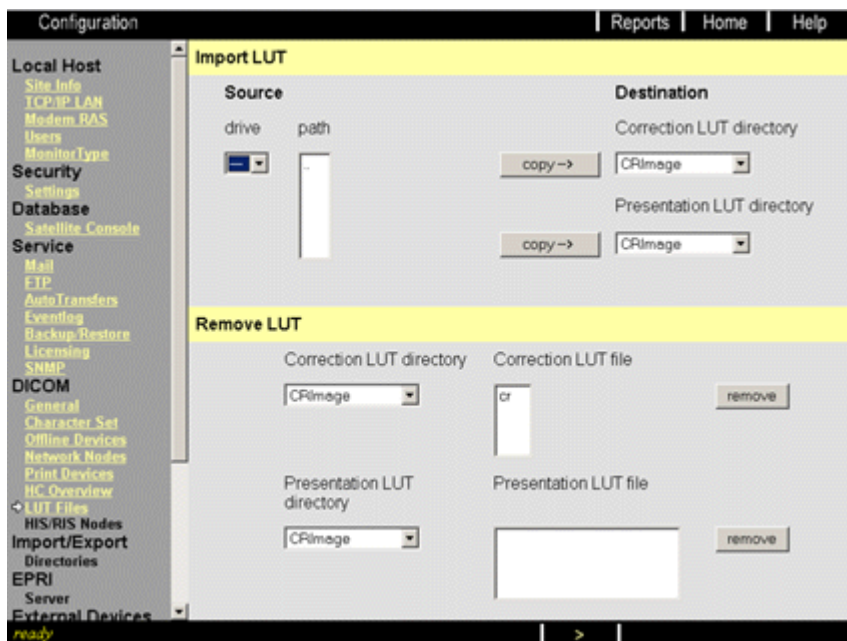


Fig. 46: DICOM LUT Files menu (import / remove)

Import LUT

If new LUT files for one or more modalities are available on a removable medium, e.g., floppy or CD, these files may be selected on the source drive and copied to the dependent LUT directories on the local system.

- **Source**

Select the drive and path where the appropriate LUT file is located.

- **Destination**

Select the destination LUT directory, depending on the image type. Click on the **Copy** button to transfer the selected LUT file.

Remove LUT

LUT files that are not needed may be removed from LUT directories on the local host by selecting the appropriate LUT file and clicking the **Remove** button.

Step 2a: Set LUT files - Correction LUT

This setting is used when the DICOM camera is set to a linear LUT and is not calibrated according to the "Gray Scale Standard Display Function" defined in the DICOM Standard (**Barten** curve).

The modality-specific LUTs (now called Correction LUTs) are used for diagnostics and for correcting the camera-specific optical density values. It has no effect on the monitor.

There may be 8-bit LUTs and 16-bit LUTs in the LUT selection box. This does not mean that one kind of LUT is made for 8-bit images and the other for 16-bit images. It means that the 8-bit LUTs have only 256 values to describe the LUT and the 16-bit LUTs have 65 kbits and are therefore much more precise.

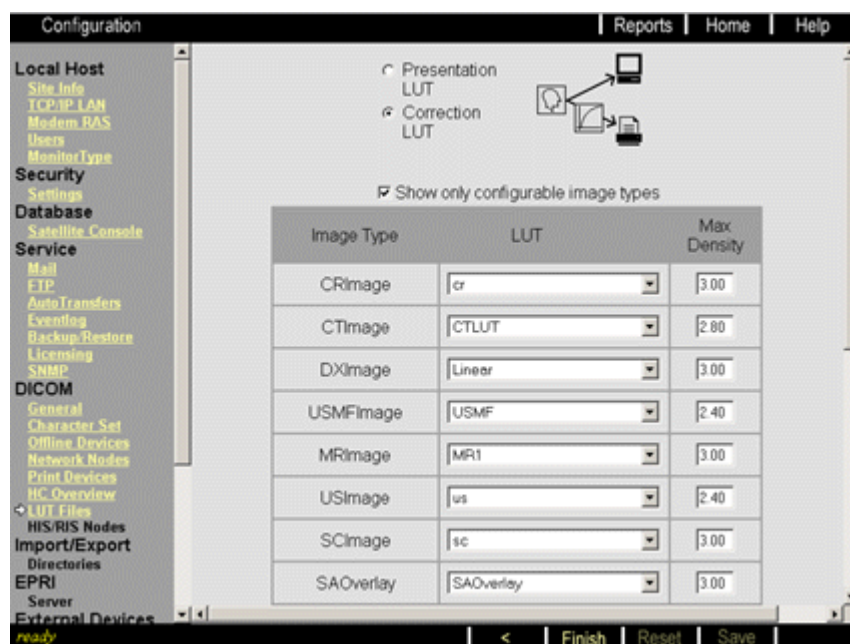


Fig. 47: LUT Files - Correction LUT enabled

Recommended settings for syngo MR systems: Select Correction LUT

	MRImage	SCImage	Your HC Device (Camera)
LUT	mr1Corr16b.lut	mr1Corr16b.lut	No LUT (Linear)
Max Density	3.0	3.0	

- **Presentation LUT**

The Presentation LUT is a camera-independent, modality-specific LUT which assumes that the camera is calibrated according to the **Barten** curve specified in the DICOM Standard, Section 14 - Gray Scale Standard Display Functions.

- **Correction LUT**

The modality-specific 'Correction LUT' needs to be selected if the camera is not calibrated according to the DICOM standard but according to a linear LUT.

- **Show only configurable image types**

Only image types in which Look up tables are loaded (imported) are shown.

- **Max. Density**

The Max. Density value needs to be set for each LUT. Usually it is not necessary to change these values, since they have already been verified by the lab and imported together with the LUT. The Min. Density value is set once in the Print Device menu.

- **Overall Maximum Density**

If images from several modalities are used in the virtual film sheet, this overall max. density value is used to recalculate the image presentation for each image.

- **Interpolation for Printing**

The entire film sheet is calculated in the local system and then sent in the page mode to the camera. To fit the images into the film sheet, pixel interpolation is necessary. The optimum value has already been evaluated at the laboratory. For this reason, modification of this parameter is not necessary. This setting has no effect on the monitor.

The following interpolation algorithms are possible:

Replicate: The interpolated value is identical to the last pixel value.

Bilinear: The interpolated value (linear) is between the last pixel value and the next pixel value

Cubic: The interpolated value (curve) is between the last pixel value and the next pixel value.

No Magnification: Leaves the image as is.

- **Mode**

Several modes may be selected to adjust the smoothing factor for the cubic interpolation algorithm.

Step 2b: Set LUT files - Presentation LUT enabled

In a Multi Vendor environment, it is often necessary to use the Presentation LUT defined by the DICOM Standard, Section 14 -> **Gray Scale Standard Display Function**. Since this LUT is new, it may not yet be available for certain modalities. This LUT is used for printing as well as for the monitor!

There may be 8-bit LUTs and 16-bit LUTs in the LUT selection box. This does not mean that one kind of LUT is made for 8-bit images and the other for 16-bit images. It means that the 8-bit LUTs have only 256 values to describe the LUT and the 16-bit LUTs have 65 kbits and are therefore much more precise.

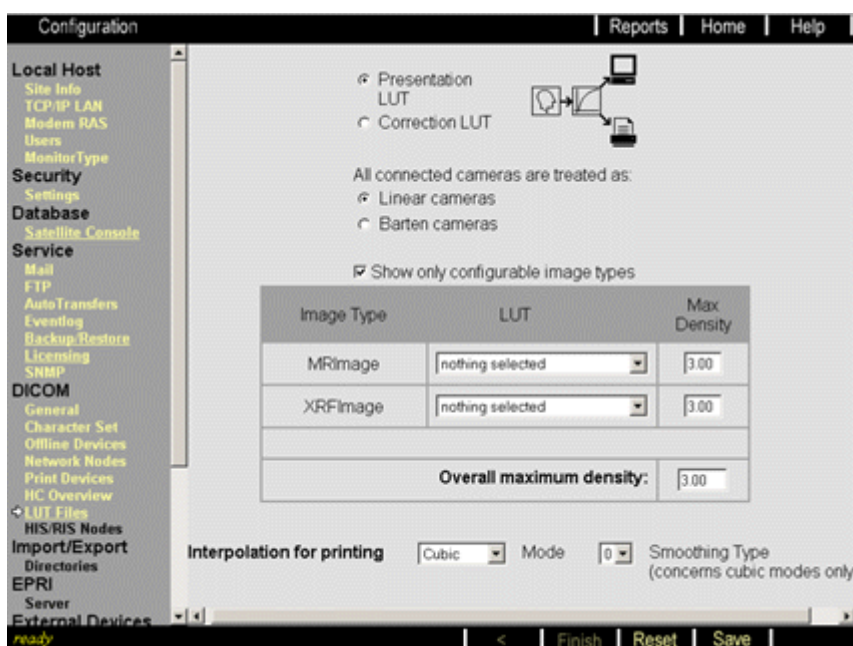


Fig. 48: LUT file - Presentation LUT enabled - Linear cameras

- **Presentation LUT**

The Presentation LUT is a camera-independent, modality-specific LUT which assumes that the camera is calibrated according to the Barten curve specified in the DICOM Standard, Section 14 - Gray Scale Standard Display Functions.

- **Correction LUT**

The modality-specific “Correction LUT” needs to be selected if the camera is not calibrated according to the DICOM standard but according to a linear LUT.

All connected cameras are treated as: (appears only if the Barten LUT is selected)

- **Linear Cameras**

The “Correction LUT” usually should be selected for cameras calibrated with a linear LUT. For certain new modalities, only a Presentation LUT exists. Therefore, the Presentation LUT needs to be selected. In addition, this button indicates that the camera does not support Presentation LUTs.

In this case, a Barten LUT is added to correct the difference between the Barten and the linear LUT.

It is not possible to mix cameras that support (and are set to) this Gray Scale Standard Display Function with cameras that do not support GSDF!

- **Barten Cameras**

In this case, all connected cameras need to support calibration according to the DICOM Standard with what is called the Barten curve, which defines luminance values as a function of the pixel value.

This calibration makes it possible to send a camera-independent Presentation LUT.

It is not possible to mix cameras that support (and are set to) this Gray Scale Standard Display Function with cameras that do not support GSDF!

- **Show only configurable image types**

Only image types in which look-up tables are loaded (imported) are shown.

- **Max. Density**

The Max. Density value needs to be set for each LUT. Usually it is not necessary to change these values, since they have already been verified and imported together with the LUT. The Min. Density value is set once in the Print Device menu.

- **Overall Maximum Density**

If images from several modalities are used in the virtual film sheet, this overall max. density value is used to recalculate the image presentation for each image.

- **Interpolation for Printing**

The entire film sheet is calculated in the local system and then sent in the page mode to the camera. To fit the images onto the film sheet, pixel interpolation is necessary. The optimum value has already been evaluated at the laboratory. For this reason, modification of this attribute is not necessary. This setting has no effect on the monitor.

The following interpolation algorithms are possible:

Replicate: The interpolated value is identical to the last pixel value.

Bilinear: The interpolated value (linear) is between the last pixel value and the next pixel value.

Cubic: The interpolated value (curve) is between the last pixel value and the next pixel value.

No Magnification: Leaves the image as is.

- **Mode**

Several modes may be selected to adjust the smoothing factor for the cubic interpolation algorithm.

HIS/RIS Nodes

Configuration of a Hospital Information System (HIS) or Radiology Information System (RIS) that supports DICOM Basic Worklist or the Detached Study Transfer Service. In many cases, the RIS is connected via an interface box such as the Mitra Broker (MagicLink HL7).

Required Data

It is the responsibility of the Project Manager to obtain the required data from the customer.

- Host name and TCP/IP address of the remote host.
- Application Entity Title (AET) and port number of the corresponding DICOM processes on the remote host
- Usually the camera has to be configured as well. It is recommended that the service engineer from the printer vendor be on-site at the same time.

Step 1: TCP/IP Settings for HIS/RIS Server

Fig. 49: DICOM HIS / RIS Nodes menu - step 1

Select Host

Click on the “Select Host” drop-down menu to see whether the HIS/RIS node is already listed. If so, and if the related data are correct, you are done and may continue with the next menu.

Host Properties

- **Host Name**

Enter the host name of the HIS/RIS server.

NOTE**Restrictions for the Host Name:**

The host name may contain up to 15 characters. The characters “_” and “-” should be avoided, since they could cause problems (even though it is possible to enter them). The computer name may not consist of just numeric characters or start with a numeric character.

- **Find**

Select the **Find** button to check whether the host name is known at the WINS or Domain Name Server (DNS). If the name is found, the IP address is entered in the TCP/IP address box.

- **TCP/IP Address**

If no DNS is used, you need to manually enter the IP address into these boxes.

- **Test**

Click on the **Test** button to send a PING command to the corresponding IP address. A TCP/IP ping may fail for the following reasons:

- The remote host is turned off or not in the same network (or subnet), or the gateway is not configured.
- The TCP/IP address is configured incorrectly.
- Any networking device (router, switch, bridge,...) denies access to the other network.

Connected by LAN / RAS

For performance reasons, an HIS/RIS should not be connected via RAS.

Step 2: DICOM Settings for HIS/RIS Server

Fig. 50: DICOM - HIS / RIS Nodes menu

General Node Properties

- **Logical Name**

Enter a logical name for the HIS/RIS node. Since the DICOM worklist requests are performed in the background and only one HIS/RIS is selectable, the logical name is not displayed in the customer user interface.

- **Edit Name**

Create or update a logical name for a HIS/RIS node.

- **Host**

Select the host name (computer name) of the HIS/RIS server.

Application Entity Properties

- **AE Title**

Select the Application Entity Title for the HIS/RIS or define a new one in the “Rdit AE Title” box. The AETs already configured are shown in this pull-down menu.

NOTE

Restrictions for the AET:

- Maximum 16 characters are allowed
- BLANKS and " (double quotes) are not allowed
- You can use alphabetic and numeric characters as well as “ _ “ and “_“.
- It is recommended to write AETs in upper case.

- **Edit AE Title**

Enter the Application Entity Title of the HIS/RIS server and click on the **Add** button.

- **Port Number**

Enter the port number of the HIS/RIS server and click on the **Add** button.

The port number is the TCP/IP port where the DICOM process (defined by the AET) in the HIS/RIS server is listening.

- **Add / Del**

If a previously configured AET is selected, the **Add** button changes to the **Del** button, allowing you to delete this AET.

- **Verification**

DICOM verification is a DICOM service for test purposes. What is called a C-ECHO command is transferred to the selected DICOM service, which is defined by the IP address, the port number, and the AET.

A C-ECHO may fail for the following reasons:

- TCP/IP address, AET, or port number is not configured correctly.
- The DICOM process at the remote host is not running.
- Certain products check the IP address or AET of the sending system. If the AET or IP address of the local system is not entered correctly there, DICOM verification will fail.
- The remote host does not support DICOM Verification as a Service Class Provider (see DICOM Conformance Statement).

Supported DICOM Services

Basic Worklist

Basic Worklist is the DICOM Service **Modality Worklist** and is used to retrieve patient and study information from the Information System (e.g. RIS).

- **Query Waiting Time**

This is the timeout of the DICOM Worklist process waiting for a response from the HIS/RIS server.

- **Max. Query Match Number**

This parameter is used to limit the number of worklist entries to be transferred.

- **Query Interval**

Enter the time in minutes between DICOM Worklist requests (polling).

NOTE

To activate the automatic DICOM Worklist polling mechanism, it is also necessary to set 'HIS/RIS update Worklist in Background' in the customer user interface under 'Option - Configuration - Patient Registration'.

- **Automatic Removal of Cancelled\Rescheduled Requests**

If an examination was sent in a previous worklist and is therefore entered in the Scheduler Database, it is removed automatically if it is no longer sent in the worklist.

Performed Procedure Step

Prior to and after patient acquisition, information data is sent to the Information System (e.g. RIS) by the DICOM Service "**Modality Performed Procedure Step**". Enable this checkbox if the selected Information System is the destination for this DICOM service.

- **Default Catalog Name**

MPPS uses codes instead of readable values. Enter the name of the coding catalog used in the Information System.

- **Default Catalog Version**

MPPS uses codes instead of readable values. Enter the version of the coding catalog used in the Information System.